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RESEARCH ARTICLE

Efficient XAI-Based Federated Learning Approach for Accurate Detection of False Data in Smart Grid

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ABSTRACT A major challenge in smart power grids (SGs) is the injection of false data by malicious actors, which may disrupt the proper operation and cause financial losses. To detect such false data, machine learning (ML)-based methods are commonly used, with models being trained through either centralized learning (CL) or federated learning (FL) approaches. While CL approach achieves higher classification accuracy, it introduces significant privacy risks. In contrast, the existing FL approaches mitigate privacy concerns by keeping data decentralized but suffer from lower classification accuracy compared to CT approach in addition to high overhead in terms of communication. To address these limitations, we propose an explainable artificial intelligence (XAI)-based FL approach that is trained on the explanations of electricity power consumption readings instead of the readings themselves. This approach not only improves false data detection accuracy but also significantly reduces FL communication overhead. The idea is that by leveraging XAI, our approach can effectively distinguish between benign and malicious samples, enabling the establishment of a precise decision boundary with fewer training rounds. To verify our approach, we conducted extensive evaluations using two real-world -electricity consumption datasets under various cyber-attack scenarios. The experimental results demonstrate that our XAI-based FL approach outperforms the existing data-based FL approaches, achieving superior accuracy and reduced overhead. For instance, in some cases, our approach was able to reduce the number of FL rounds by more than 80%, reduce the false alarm by more than 82%, and increase the detection accuracy by more than 14%.

INDEX TERMS False data detection, federated learning, explainable artificial intelligence, privacy preservation, electricity theft, smart grid.

I. INTRODUCTION

A. BACKGROUND AND LITERATURE LIMITATIONS

Smart power grids (SGs) offer numerous advantages over traditional power grids by enabling real-time data exchange between electricity providers and consumers. This improved grid can boost energy efficiency, enhance service reliability, and support the integration of renewable energy sources [1], [2]. With this capability, smart grids can quickly adapt to changes in energy demand and respond to outages in real-time. However, they are vulnerable to cyber-attacks, particularly targeting advanced metering infrastructures (AMIs).

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These attacks manipulate data sent by smart meters (SMs), leading to injecting false power consumption readings [3]. These false readings are usually lower than the actual ones to reduce electricity bills and this attack is called energy theft (ET).

The injection of false readings in the smart grid disrupts load monitoring and energy management systems, causing both operational inefficiencies and substantial financial losses for electricity providers [4]. To detect these false readings, machine learning (ML)-based detectors are utilized, analyzing consumption patterns to detect anomalies. This consumption-pattern-based method is more cost-effective than sensor-based approaches, which require additional equipment and sensors, leading to high maintenance and operational costs [5].

ML-based detectors are trained using either centralized learning (CL) or federated learning (FL) approaches [6]. CL approach raises privacy concerns as it requires sharing detailed consumer data, which could reveal personal information like daily routines, appliance usage, and presence at home, potentially leading to malicious exploitation [7]. Therefore, FL addresses these concerns by storing data locally at consumer side and only sharing model updates with a central server. However, FL detectors generally perform worse than CL detectors in terms of detection accuracy, particularly in non-independent and identically distributed (Non-IID) data scenario [8], [9], [10], [11]. Moreover, the existing FL approaches suffer from high communication overhead during training process. This can be identified as Limitation #1 in the literature, i.e., detectors performance and overhead challenges in FL.

On the other hand, explainable artificial intelligence (XAI) focuses on making the decisions of ML black-box detectors transparent and understandable [12], [13]. By providing clear explanations for how models make their classifications, XAI aims to build trust and enhance reliability [14], [15]. Therefore, most XAI research in the context of FL focuses on post-hoc explanations, meaning interpretability is addressed only after the training process is complete. While XAI explanations can effectively distinguish between data from different class labels, there is a lack of research in the literature on utilizing XAI techniques to enhance the performance of FL training process, particularly with regard to improving classification accuracy and reducing overhead. This can be identified as Limitation #2 in the literature, i.e., lack of investigation on the use of XAI in FL training process to enhance performance.

B. CONTRIBUTIONS

This paper aims to address the two main Limitations (#1 and #2) in the literature discussed earlier. To the best of our knowledge, this is the first work to propose an enhanced FL approach that leverages training on XAI instead of raw data to improve performance and reduce overhead.

The main contributions of our work are as follows:

- **We propose an XAI-based FL false data detector to enhance performance and reduce overhead.**
  - This detector is a deep ML-based model trained on XAI-generated explanations of power consumption readings, rather than the raw readings themselves. This approach enhances classification accuracy and reduces overhead. This is because XAI effectively distinguishes between benign and malicious samples, making it easier for the ML detector to establish decision boundaries. As a result, the boundaries are less complex, which in turn requires fewer communication rounds during training, further reducing overhead.
- **We have conducted extensive experiments to evaluate the proposed XAI-based FL false data detector and compare it to the existing data-based FL false data detector against well-known false data injection cyber-attacks, considering both IID and Non-IID scenarios.**
  - The experimental results demonstrate that the proposed XAI-based FL detector outperforms the data-based FL detector in both accuracy and overhead reduction.
- **We have assessed our proposed XAI-based FL detector across different real-world electricity consumption data sources.**
  - The proposed FL approach (i.e., utilizing XAI) demonstrates its applicability and effectiveness on both the Irish Smart Energy Trial (ISET) dataset [16] and the Pakistan Residential Electricity Consumption (PRECON) dataset [17].

C. PAPER ORGANIZATION

The structure of this paper is as follows. Section II presents a review of prior work on FL-based false data detection and the integration of XAI within FL. The proposed XAI-based FL approach is explained in section III. Section IV introduces the datasets used in the experiments and false data injection attacks. Section V presents the experimental setup and evaluations. Finally, section VI concludes the paper and provides future works.

II. RELATED WORKS

Several studies in the literature highlighted the growing role of machine learning in smart grid applications, energy systems, and cybersecurity. In load data restoration, a Generative Adversarial Networks (GAN) based model, Load Profile Inpainting Network (Load-PIN), effectively reconstructs missing energy profiles and outperforms existing deep learning methods on real-world datasets [18]. For power grid stability analysis, a hybrid neural Lyapunov method combines linear and nonlinear models to reduce conservatism and improve validation efficiency, achieving more reliable estimates of the domain of attraction [19]. In ensemble learning, a novel approach called func-bagging integrates adaptive weight distribution into heterogeneous classifiers,

outperforming traditional bagging and stacking techniques in class-imbalanced tasks [20]. In blockchain security, a deep learning method (NDLSC) using opcode-level analysis and combining Residual Networks (ResNet) and Kolmogorov-Arnold networks (KAN) models improves smart contract vulnerability detection accuracy and robustness [21]. Finally, a hybrid Multi-Layer Perceptron (MLP) and Gated Recurrent Unit (GRU) model addresses data imbalance and enhances electricity theft detection accuracy when applied to real data from the Chinese National Grid Corporation [22].

Several works have studied the threat of false data injection attacks in power systems and non-FL environment. For instance, the study in [23] introduces a mathematical model of false data injection attacks based on incomplete topology and AC state estimation, aiming to bypass limitations of DC-based attack models. A graph neural network-based localization approach is also developed, enabling accurate, robust, and adaptive detection of false data injection attacks across varying grid topologies.

In this section, we first review the state-of-the-art FL false data detection, and then we discuss the use of XAI to improve the transparency and interpretability of FL. Finally, we highlight the research gaps in the literature that this paper aims to address.

A. FL FALSE DATA DETECTION

Existing methods for detecting false data in FL can be categorized into supervised [24], [25], [26], [27] and unsupervised [28], [29], [30], [31] approaches. Supervised methods train binary detectors using both benign and malicious data to distinguish between the two patterns, while unsupervised methods use only benign data to train anomaly detectors, which aim to identify irregular patterns.

1) SUPERVISED FL

Wen et. al. [24] proposed a privacy-preserving FL framework that includes data center, control center, and multiple detection stations. The proposed framework utilizes a temporal convolutional network deep learning model to effectively detect ET false data in SGs. It also applies a perturbation technique to the consumption data, utilizing a simple local differential privacy scheme to preserve consumer privacy against the detection stations. Although the proposed framework can achieve better performance compared to the existing approaches, it incurs additional computational overhead. Similarly, Ashraf et. al. [25], proposed an FL framework that utilizes a federated voting classifier to detect false data causing ET in SGs, while ensuring the security of client data. This framework includes random forests, k-nearest neighbors, and bagging classifiers as an ensemble of models based on majority voting.

In [26], Ibrahem et. al. proposed a lightweight, privacy-preserving FL using a functional encryption cryptosystem to detect ET false data. This method eliminates the need for a key distribution center by allowing detection stations to send encrypted local model parameters to an aggregator, securing them against membership and inference privacy attacks. The experiments demonstrate that the proposed functional encryption cryptosystem is more efficient than the existing cryptosystems [24]. Moreover, Wang et. al. [27] proposed a decentralized FL framework using threshold homomorphic encryption for parameter aggregation between distribution system operators, eliminating the need for a federated server. The experiments show that the extreme gradient boosting FL detector outperforms the multilayer perceptron FL detector in both IID and non-IID datasets.

The paper in [32] proposes a privacy-preserving method for detecting false data injection attacks in smart grids by integrating transformers, federated learning, and the Paillier cryptosystem. The transformer model captures complex relationships in electrical signals using its self-attention mechanism, while the Paillier encryption enhances security during training. Experimental results on IEEE 14-bus and 118-bus systems confirm the effectiveness and robustness of the proposed approach.

The paper in [33] presents a comprehensive survey on the role of FL in enhancing the security and privacy of future 6G networks. The paper reviews the background of 5G and FL, explores current FL applications in 5G, analyzes key threats and vulnerabilities in FL systems, and discusses how advanced FL techniques can address emerging security challenges in 6G environments.

2) UNSUPERVISED FL

Mohammadi et. al. [28] proposed an unsupervised FL anomaly detection method. They explore the use of a long short-term memory autoencoder (LSTM-AE) and its integration with one-class support vector machines and isolation forest. The experimental results show that threshold-based techniques are more effective at detecting abnormalities, and the proposed method outperforms others when evaluated on a synthetic dataset. Similarly, Shrestha et. al. [29] proposed an FL anomaly detection method based on the LSTM-AE model, using mean standard deviation (MSD) and median absolute deviation (MAD). The experimental results show that the MSD approach outperforms the MAD approach and highlight a trade-off between privacy, performance, and computation time when using encryption. Specifically, preserving privacy requires a compromise in execution time and performance.

In [30], Jithish et. al. proposed an FL detector designed to ensure user privacy and secure model parameter updates through a secure communication protocol. While their work highlighted the ability of FL to ensure user privacy, it also revealed significant resource constraints, including increased computation, time, and communication overhead at edge devices with limited memory and processing capacity. Additionally, Alshehri et. al. [31] proposed a privacy-preserving FL framework for deep anomaly detection of false data associated with ET in SGs. In this framework, consumers

train local detectors on their private electricity usage data and share only the trained detector parameters with an electric utility company's aggregation server to collaboratively build a global anomaly detector. The experimental results show that although the proposed detector ensures privacy, CL approach outperformed it in detection accuracy, highlighting the trade-off between privacy and accuracy.

B. INTERPRETABLE FEDERATED LEARNING

Research on interpretable FL models can generally be divided into post-hoc and pre-hoc approaches, which differ in when and how interpretability is incorporated. Post-hoc methods, such as [15], [34], [35], [36], [37], and [38], analyze a trained model treated as a black box, applying external techniques to explain its predictions without altering the underlying architecture. This makes them highly adaptable and suitable for complex models. In contrast, pre-hoc methods, such as [39], [40], and [41], design interpretability into the model itself by using transparent structures such as decision trees, rule-based systems, or fuzzy logic models. These inherently interpretable models provide direct, exact explanations but may trade off some predictive performance or scalability compared to high-capacity black-box models.

1) FL POST-HOC XAI

Huong et. al. [15] presented a FL anomaly detection designed for Internet of Things (IoT)-based industrial control systems in smart manufacturing. They integrated XAI to provide interpretability, enhancing trust and decision-making by experts in real-time scenarios. Similarly, Attique et. al. [34] introduced a FL intrusion detection system designed to secure industrial IoT infrastructures while preserving data privacy. The proposed system uses a deep neural network in a federated setting to detect anomalies in industrial IoT traffic, addressing the increasing security threats in industrial environments. To enhance interpretability, they used Shapley Additive Explanations (SHAP), which provides insights into the model's decision-making process. Additionally, in [35], the authors proposed an XAI FL approach to address the challenges of detecting fraudulent transactions, such as the rarity of such transactions, the imbalanced nature of transaction datasets, and data privacy laws that prevent institutions from sharing customer data for centralized models. By adopting XAI, the approach enhances transparency, making the model's predictions interpretable to human experts. This interpretability fosters trust, complementing the privacy-preserving capabilities achieved by FL.

Wang et. al. [36] explored how to fairly assess the contributions of multiple participants in FL using XAI. They proposed a method for calculating contributions in both horizontal and vertical FL scenarios using shapley values. The experimental results show that the proposed method is effective, providing fair and reliable contribution measurements for FL participants while maintaining data privacy. Additionally, the authors of [37] proposed a FL approach addressing the challenge of relevant client selection, which involves identifying clients with relevant data to meet the central server's learning objectives. They applied XAI using the shapley value concept from cooperative game theory, formulating a cooperative game based on gradients shared by clients. This led to the development of the shapley value-based federated averaging algorithm, which improves the server's ability to effectively select relevant clients. Their empirical analysis on image classification and speech recognition tasks shows that the proposed approach outperforms existing baseline methods in supervised FL scenarios, demonstrating its effectiveness in enhancing client selection in FL.

Similarly, Wang et. al. [38] proposed a method for estimating shapley values in FL without requiring model retraining, addressing a major limitation of traditional shapley value assessments used for tasks like participant selection and reward allocation. The method is applicable to both vertical and horizontal FL. The empirical evaluations on 14 public datasets demonstrate that the estimated shapley values closely match actual values. Additionally, the proposed method outperforms existing approaches especially in scenarios where more than 80% of participants contribute low-quality data.

2) FL PRE-HOC XAI

Chen et. al. [39] proposed a vertical FL framework for credit scoring in financial decision-making, utilizing gradient boosting decision trees and random forests due to their interpretability and efficiency. Their analysis and experimental results demonstrate that the framework enhances model interpretability while maintaining efficiency and accuracy. Similarly, the authors of [40] introduced a tree-based FL using a histogram-sharing scheme that supports both horizontal and vertical FL of gradient boosting decision trees with configurable privacy protections. Additionally, Bárcena et. al. [41] proposed a FL approach using fuzzy rule-based models, which are considered XAI models suitable for regression tasks. In this approach, independent data owners locally create fuzzy rule models following a standardized procedure. These local models are sent to a central server, which aggregates them into a global model using a strategy designed to preserve explainability. The experimental results, evaluated using mean squared error across four regression datasets, demonstrate that the FL approach improves accuracy compared to local models. However, it is outperformed by the CL approach, which delivers better performance but they are unsuitable for privacy-sensitive scenarios.

Unlike [15], [34], [35], [36], [37], [38], [39], [40], [41], which aim to design interpretable FL using either Post-Hoc or Pre-Hoc XAI methods, this paper focuses on proposing an efficient FL approach that enhances classification accuracy while reducing overhead through XAI. The proposed XAI-FL approach is trained on XAI-generated explanations of power consumption readings, rather than the raw readings themselves. These explanations are produced using a

Pre-Hoc method, where each FL consumer participant uses a pre-trained random forest model to generate SHAP values locally. These explanation-based features are then used during the FL training process.

C. RESEARCH GAPS

As outlined in the previous two subsections, existing studies on FL-based false data detectors show less performance compared to CL-based methods, reflecting a trade-off between privacy and detection accuracy in FL. Moreover, most FL studies that incorporate XAI in the FL training process primarily focus on enhancing model explainability, without exploring using it to enhance classification accuracy and reduce overhead. Therefore, several research gaps remain unexplored as follows:

- **Lower FL performance.** While FL-based false data detectors preserve consumer privacy compared to CL-based methods, their performance is outperformed by CL-based methods in terms of classification accuracy, making FL-based approaches less practical for real-world use. Therefore, achieving both privacy preservation and high classification accuracy remains a challenge.
- **High overhead.** The existing FL approaches require high communication/computation overhead, especially after considering the encryption operations needed to hide the local model parameters from the aggregation server for privacy preservation. Therefore, exploring novel approaches to improve the overhead of the FL training process remains a challenge.

To address these challenges, this paper explores a novel FL training approach that utilizes XAI to enhance classification accuracy and reduce overhead. While the existing works focused on using XAI to provide explanations in FL to improve the understanding of decisions, we explore training the detectors on them rather than the raw data to address the aforementioned challenges. The rationale behind this approach lies in XAI's ability to clearly distinguish benign and attack patterns, which helps streamline the learning process for the ML detector. Consequently, these boundaries become less complex, reducing the number of communication rounds needed during training and thereby lowering overall overhead.

III. THE PROPOSED APPROACH

In this section, we introduce our proposed XAI-based FL training approach and provide the rationale behind it.

A. THE PROPOSED XAI-BASED FL APPROACH

The system model considered in this paper operates between two parties: the electric utility company (EUC) and consumers, who have SMs installed at their premises. The SMs periodically record fine-grained power consumption readings for each consumer. These readings are used by the EUC for various purposes, such as training a binary classifier for the detection of false power consumption readings. Binary classifiers are machine learning models designed to distinguish between two classes, typically labeled as positive and negative (e.g., 0 and 1). They learn from labeled training data by identifying patterns or decision boundaries that separate the two classes. During inference, the classifier takes an input sample and predicts its class based on the learned patterns. However, consumers are unwilling to share these detailed power consumption readings directly with the EUC for building a CL detector because doing so could compromise their privacy and expose sensitive information. Conversely, building a FL detector may lead to reduced performance and impose high overhead. To address this concern, the proposed XAI-based FL approach is designed to preserve privacy while maintaining high performance and minimizing overhead.

Additionally, we select a CNN as our classifier, as CNNs—while traditionally used for image processing—have demonstrated strong effectiveness in analyzing time series data, such as power consumption patterns. Their hierarchical structure enables them to learn both localized and sequential features by applying filters across time steps, allowing them to capture short-term trends and subtle variations in the data. This makes CNNs particularly suitable for identifying repetitive behaviors or sudden changes in temporal signals. Moreover, CNNs minimize the need for manual feature extraction by learning meaningful patterns directly from raw input. Their support for parallel computation and ability to handle multichannel data further enhance their suitability for real-time and high-dimensional time series classification tasks.

The training and inference phases of the proposed XAI-based FL approach are detailed as follows:

- **Training phase**: Fig. 1 illustrates this phase, where N consumers are selected to participate in building the global model. These consumers work with the EUC to execute our XAI-based FL approach. These participating consumers represent a subset of all consumers connected to the EUC. They train local models locally on XAI representations (explanations) of their power consumption readings, rather than directly using the raw readings collected by their SMs. To extract these XAI representations, each consumer applies SHAP to a pre-trained random forest model trained on their local power consumption data. Once the random forest model is trained, the TreeSHAP method is employed to generate interpretable feature attributions that explain how each input contributed to the model's predictions. SHAP assigns a contribution value to each feature (power consumption reading) based on cooperative game theory, indicating the degree each feature (reading) influenced the pre-trained model's predictions [42]. TreeSHAP offers a more efficient alternative to traditional SHAP methods [43], [44]. While classical SHAP approaches require evaluating all

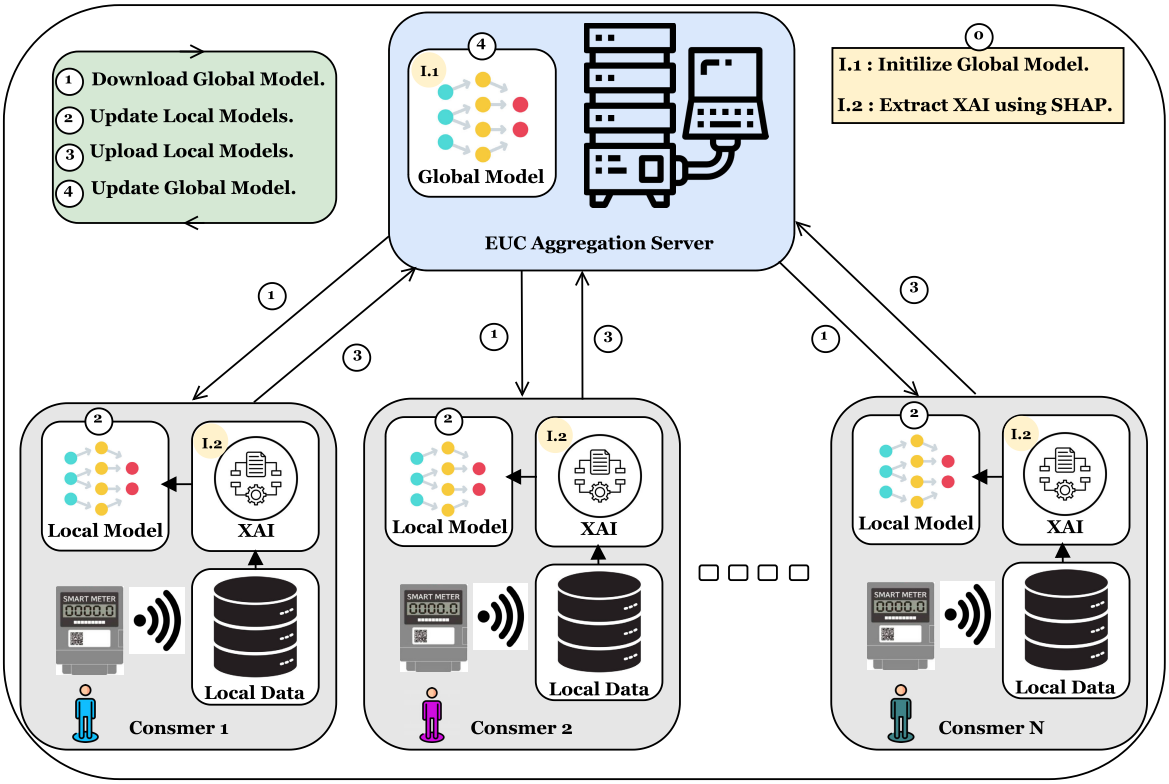


FIGURE 1. The proposed XAI-FL approach (Training phase).

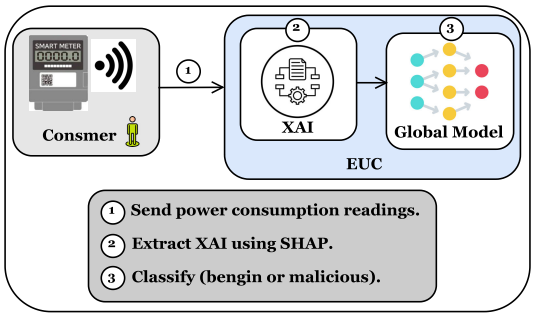


FIGURE 2. The proposed XAI-FL approach (Inference phase).

possible feature combinations to compute exact Shapley values—an operation that becomes computationally expensive as the number of features increases—TreeSHAP leverages the structure of decision trees to calculate exact SHAP values in polynomial time. By exploiting the recursive nature of tree splits and paths, TreeSHAP avoids the need for background sampling or enumeration of feature subsets. This results in a significant reduction in computational overhead, enabling fast and accurate explanation generation without compromising interpretability, even for large datasets or complex ensembles.

The EUC employs an aggregation server to combine these locally trained models into a global model without requiring the sharing of raw data; only model gradient updates are exchanged between the two parties. The model is trained to detect ET over an entire day, using the power consumption readings collected from the SM throughout the day. The detailed interactions between the EUC aggregation server and consumers are summarized in Algorithm 1.

- **Inference phase**: Fig. 2 illustrates this phase, which involves a data communication process between the EUC and one consumer, and this process is the same for all consumers connected to the EUC. This phase consists of the following steps. First, the power consumption readings are sent from a consumer's SM to the EUC. Second, the EUC extracts XAI representations (explanations) of these readings by applying SHAP to a pre-trained model that was trained on a historical public dataset. Third, the EUC's global model classifies these readings as either benign or malicious.

B. THE RATIONALE BEHIND THE PROPOSED XAI-BASED FL

1) FL VS CL

Machine learning models for detecting false power usage can be trained in centralized or federated modes. While centralized learning offers simplicity and centralized control, it involves collecting detailed consumption data at a central

Algorithm 1 The Proposed XAI-FL System Model

```
/* Step 0: Initialization */
The EUC aggregation server initializes the global detector parameters.
Consumers extract XAI features from their power consumption data using the SHAP algorithm.
for each round in FL rounds do
    /* Step 1: Download Global Model */
    The EUC aggregation server distributes Global Model to all consumers.
    /* Step 2: Update Local Models */
    Consumers train a local model using the XAI features of their power consumption data.
    /* Step 3: Upload Local Models */
    Consumers send their updated local model parameters to the EUC aggregation server.
    /* Step 4: Update Global Model */
    The EUC aggregation server updates the global model by averaging the parameters received from consumers.
end
```

server, which raises significant privacy concerns. On the other hand, federated learning maintains user confidentiality by ensuring that raw data never leaves local devices. Instead, participants update models locally and share only model parameters, reducing the risk of sensitive data exposure.

2) DATA-BASED FL VS XAI-BASED FL

Federated learning implementations can vary based on the type of input used for training. Conventional approaches rely on raw consumption records, which are often noisy and complex, making it difficult for the model to learn consistent patterns—especially under non-IID data distributions. This not only slows training but may also compromise detection reliability.

The proposed method takes a different path by introducing explanation-based inputs using SHAP-derived features. These features—obtained locally from a pre-trained decision tree model—highlight the most influential readings contributing to classification decisions. Fig. 3 presents SHAP force plots for two randomly selected samples—one benign and one malicious. Each plot illustrates 48 SHAP values, corresponding to the contribution of individual half-hourly power consumption readings across a 24-hour period. This abstraction simplifies the input representation and enables the global model to converge faster and generalize better across participants.

To assess the effectiveness of these SHAP features, we applied Principal Component Analysis (PCA) on both raw and explanation-based data under IID and Non-IID conditions. Figs. 4 and 5 show the first two principal components, PC1 and PC2, plotted along the x- and y-axes, respectively. As shown in these figures, raw input often results in overlapping clusters between benign and attack samples, especially when the adversary uses stealthy manipulation by reducing reported usage slightly (e.g., varying $\alpha < 1$). By contrast, XAI features exhibit better separation between benign and malicious patterns—even under more subtle attack settings. The dataset used in this comparison is ISET [16], and the analysis is based on attack #1, where malicious clients under-report consumption by multiplying real values with α. Additional details about this attack are provided in the subsequent section.

Figs. 4 and 5 show a noticeable overlap between benign and malicious consumption patterns when using raw data, whereas XAI features can clearly distinguish between them. It can also be seen that as the complexity and stealthiness of the attack increase (i.e., as α increases), the overlap between benign and malicious patterns also increases. However, XAI continues to effectively separate benign and malicious patterns. Only these figures are included here due to space constraints, but the same conclusions hold true across different false data injection attacks and datasets which will be discussed in next section.

One important advantage of using SHAP values in a federated context is their consistency. Since all clients use the same model architecture and follow a consistent method for computing SHAP values under a shared supervised machine learning objective (e.g., detecting the same types of attacks), their representations reflect comparable feature importance—especially when clients share similar usage behaviors. As a result, when clients exhibit similar data patterns or feature distributions (e.g., similar power consumption behavior), their SHAP explanations—representing the importance of input features—tend to be closely aligned. This is analogous to a common phenomenon in computer vision, where models trained on similar images often learn the same features, such as edges or textures.

IV. DATASETS AND FALSE DATA ATTACKS

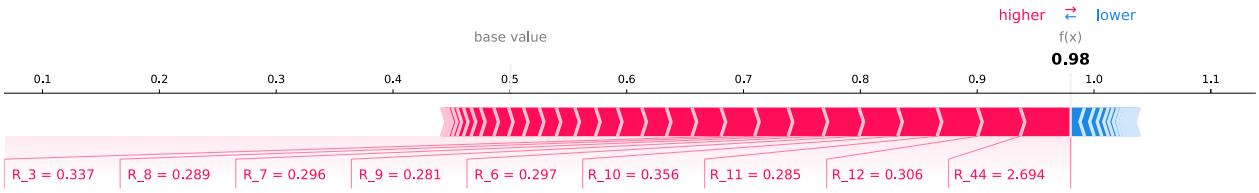
This section begins with a description of the datasets used in our experiments and then discusses the cyber-attacks targeting power consumption data.

A. DATASETS

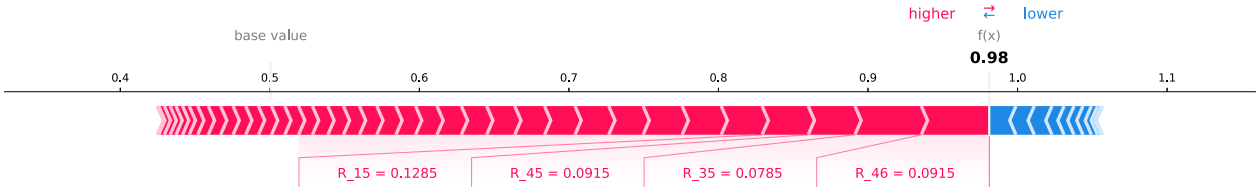
Table 1 summarizes the key characteristics of two public and real-world electricity consumption datasets. In this subsection, we describe them in details.

1) ISET DATASET

The Irish Smart Energy Trial (ISET) dataset [16] was created by Electric Ireland and the Sustainable Energy Authority of Ireland. As shown in Table 1, it was collected between 2009 and 2010 and includes benign electricity consumption readings from 3,639 residential consumers ($\mathcal{C}$) over a period of 536 days ($\mathcal{D}$). The dataset recorded 48 readings per day at half-hourly intervals (every 30 minutes).

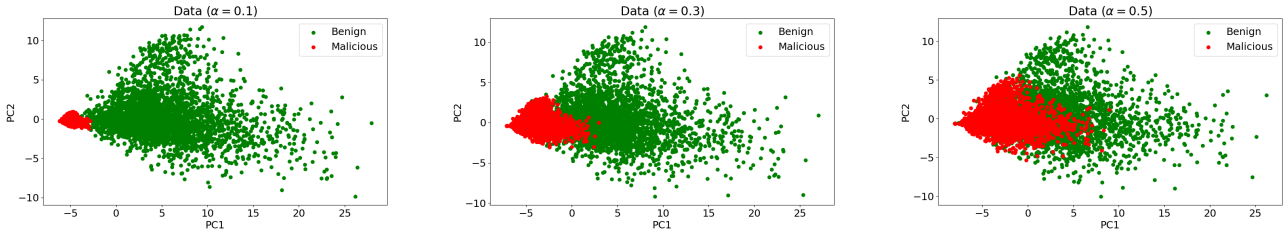


(a) Benign sample.

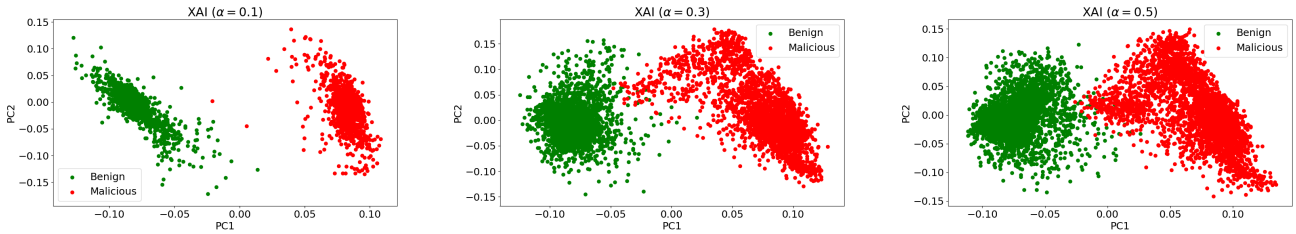


(b) Malicious sample.

FIGURE 3. SHAP feature explanations of two samples (benign and malicious).



(a) Benign and malicious Data under varying values of α for attack #1.



(b) Benign and malicious XAI under varying values of α for attack #1.

FIGURE 4. PCA applied to Data and XAI explanations of power consumption readings (IID Scenario).

2) PRECON DATASET

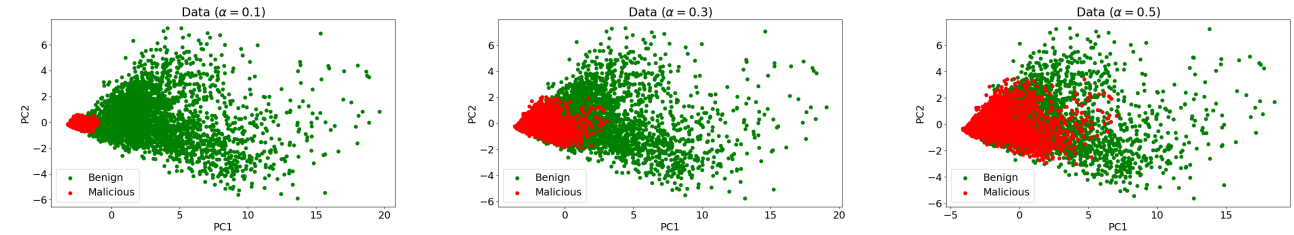
The Pakistan Residential Electricity Consumption (PRECON) dataset [17] was collected by Energy Informatics Group in Lahore University. As shown in Table 1, it was collected between 2018 and 2019 and includes benign electricity consumption readings from 42 residential consumers ($\mathcal{C}$) over a period of 321 days ($\mathcal{D}$). The dataset recorded 1,440 readings per day at one-minute intervals.

The two datasets are sourced from benign residential consumers. However, during the collection process, there may be issues such as incorrect readings, missing data, and unintended anomalies (outliers), often resulting from equipment malfunctions, inaccuracies, or transmission issues [45]. To address these potential problems, we apply data cleaning techniques to ensure the datasets are free of such issues [46].

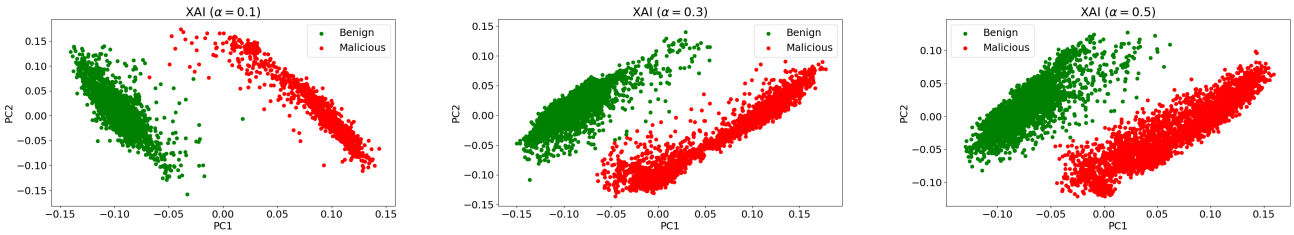
TABLE 1. Description of datasets.

	ISET	PRECON
Year range	2009-2010	2018-2019
Consumers ($\mathcal{C}$)	3,639	42
Total days ($\mathcal{D}$)	536	321
Readings per day	48	1440
Readings interval	30 minutes	1 minute

After cleaning, the datasets are used in our experiments for detecting intentional false data injection cyber-attacks,



(a) Benign and malicious Data under varying values of α for attack #1.



(b) Benign and malicious XAI under varying values of α for attack #1.

FIGURE 5. PCA applied to Data and XAI explanations of power consumption readings (Non-IID Scenario).

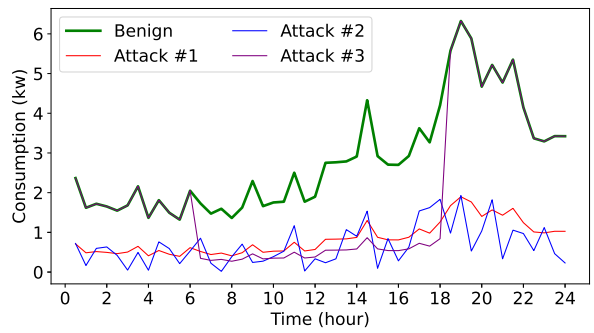
which are detailed in the following subsection. Additionally, to explore both IID and Non-IID data scenarios, we used the K-means algorithm to cluster consumers' power consumption data. For the IID scenario, all consumers were selected from a single cluster, ensuring similar consumption patterns among participants. In contrast, for the Non-IID scenario, one consumer was selected from each cluster, capturing diverse consumption behaviors for participation in the FL training. In the ISET dataset, we considered both IID and Non-IID scenarios. However, due to the limited number of consumers in the PRECON dataset, we considered only the Non-IID scenario.

B. FALSE DATA INJECTION ATTACKS

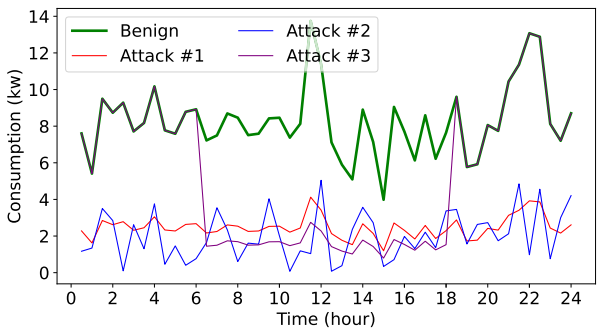
In a typical scenario where only honest consumers are assumed, the reported power consumption readings ($R_{reported}$) and the actual (or benign) power consumption readings (R_{actual}) are expected to be the same, meaning that $R_{reported} = R_{actual}$. However, cyber-attacks introduce false power consumption readings intended to mislead ML detectors. In this paper, we utilize several types of false data injection cyber-attacks to simulate scenarios where a malicious consumer alters benign (or actual) consumption readings, R_{actual} to commit ET through falsified reported consumption readings, $R_{reported}$. These attacks, frequently discussed in the literature [47], can be classified into two categories: continuous attacks, which involve sustained data manipulation, and intermittent attacks, where the manipulation occurs sporadically. In this paper, we consider the following attacks.

1) ATTACK #1

It is a continuous attack, attackers report a lower version of their actual consumption readings using a constant reduction



(a) ISET Dataset.



(b) PRECON Dataset.

FIGURE 6. Power consumption readings for one day.

factor ($0 < \alpha < 1$). The reported reading is modeled by Eq. (1). It is worth noting that a smaller value of α makes the attack more severe, as the reported consumption becomes significantly lower than the actual value. Conversely, as α increases, the attack becomes harder to detect because

the reported consumption readings closely approximate the actual values.

$$R_{repoted} = \alpha \times R_{actual} \tag{1}$$

2) ATTACK #2

It is another form of a continuous attack, where attackers report a lower version of their actual consumption readings using a time-dependent reduction factor ($0 \leq \alpha_t < 1$). The reported reading is modeled by Eq.(2). Note that attack #1 uses a constant reduction factor while attack #2 is more stealthy due to varying the reduction factor.

$$R_{repoted} = \alpha_t \times R_{actual} \tag{2}$$

3) ATTACK #3

It is an intermittent attack, where the attacker submits a lower version of their actual consumption readings using a constant reduction factor ($0 < \beta < 1$), over a specific time interval, from t_s to t_e, and submits the actual readings in the other times. The reported reading is modeled by Eq.(3). In this attack, the attackers try to confuse the detector by alternating between true and false data. It is worth noting that a larger value of β and/or a longer attack period ($t_e - t_s$) makes the attack more severe, as it results in lower reported consumption. Also, a smaller value of β and/or a shorter attack period makes the attack more stealthy, as it becomes harder to detect due to the reported consumption closely resembling the actual consumption.

$$R_{repoted} = \begin{cases} \beta \times R_{actual} & \forall t \in [t_s, t_e] \\ R_{actual} & \forall t \notin [t_s, t_e] \end{cases} \tag{3}$$

Fig. 6 illustrates the power consumption readings over a one-day period for a single consumer from both the ISET and PRECON datasets. The figure presents the benign (or actual) reading sample alongside its false (or malicious) versions, generated using the three distinct cyber-attacks (#1, #2, and #3). It can be seen that a significant reduction in electricity consumption is observed when comparing the benign sample to the malicious samples, highlighting the impact of these cyber-attacks.

V. PERFORMANCE EVALUATION

This section introduces the experimental setup and evaluation metrics, followed by a detailed discussion of the experimental results.

A. EXPERIMENTAL SETUP AND EVALUATION METRICS

We conducted our experiments on the high-performance computing cluster at Tennessee Technological University. The experiments were implemented and evaluated using various Python libraries, including pandas, Keras, Matplotlib, and scikit-learn. Both IID and non-IID data scenarios were considered, utilizing the datasets described in subsection IV-A, along with the cyber-attacks outlined in subsection IV-B, to train and test our detectors. In each scenario, 10 consumers were selected to participate in the FL training process. For each consumer, benign and malicious data were combined and then split into training and testing sets using a 2:1 ratio. Practically, the ISET dataset contains 1,072 samples (both benign and malicious) per consumer, resulting in a total of 10,720 samples across 10 consumers. Similarly, the PRECON dataset includes 730 samples per consumer, totaling 7,300 samples. Each local model was trained on the corresponding training data, while all local testing sets were merged to form a unified global testing set used to evaluate the global FL model. We trained CNN-based deep learning supervised detectors using a combination of benign and malicious data. The CNN architecture was selected because of its superior performance, which is attributed to its ability to capture patterns and correlations in input data through the filters of its convolutional layers. These filters extract essential features from input data, making CNNs particularly effective for this task.

The hyper-parameters optimization was performed to select the optimal architecture and ensure a fair comparison. The optimal hyperparameters selected for the pre-trained random forest model using Keras' `RandomForestClassifier` include n_estimators = `50`, criterion= `gini`, max_depth= `None`, and max_features= `sqrt`. Additionally, Table 2 gives the optimal architecture for the CNN-based detectors used during the FL process for both the state-of-the-art (SoTA) approach (Data-FL, which applies FedAvg directly on the dataset) and the proposed approach

TABLE 2. The optimal hyper-parameters for CNN-based detectors.

Detector	Description
CNN-based	Input, 48, Linear
	Conv1D, 128, ReLU
	Dense, 256, ReLU
	Dense, 128, ReLU
	Dense, 64, Sigmoid
	Output, 2, Softmax

TABLE 3. Performance metrics.

Metric Name	Formula
Accuracy	$ACC(\%) = \frac{TP+TN}{TP+TN+FP+FN} \times 100$
Precision Rate	$PR(\%) = \frac{TP}{TP+FP} \times 100$
F1 Score	$F1(\%) = \frac{2*PR*DR}{PR+DR} \times 100$
Detection Rate	$DR(\%) = \frac{TP}{TP+FN} \times 100$
False Alarm	$FA(\%) = \frac{FP}{FP+TN} \times 100$
Highest Difference	$HD(\%) = DR(\%) - FA(\%)$

TABLE 4. The performance comparison between the data-based and XAI-based FL against attack #1 for IID Scenario.

	$\alpha = 0.1$		$\alpha = 0.3$		$\alpha = 0.5$	
	Data-FL [24], [27], [48]	XAI-FL Proposed	Data-FL [24], [27], [48]	XAI-FL Proposed	Data-FL [24], [27], [48]	XAI-FL Proposed
ACC	99.39 ± 0.14	99.48 ± 0.17	97.28 ± 0.38	97.67 ± 0.08	88.40 ± 1.12	93.31 ± 0.30
PR	99.03 ± 0.18	99.63 ± 0.40	96.57 ± 0.39	97.86 ± 0.32	86.61 ± 0.81	93.91 ± 0.27
$F1$	99.40 ± 0.14	99.48 ± 0.17	97.30 ± 0.38	97.66 ± 0.08	88.68 ± 1.14	93.27 ± 0.23
DR	99.76 ± 0.35	99.34 ± 0.08	98.03 ± 0.51	97.47 ± 0.24	90.84 ± 1.83	92.64 ± 0.85
FA	0.98 ± 0.18	0.38 ± 0.41	3.48 ± 0.40	2.13 ± 0.33	14.04 ± 0.87	6.01 ± 0.43
HD	98.79 ± 0.27	98.96 ± 0.34	94.55 ± 0.76	95.34 ± 0.16	76.80 ± 2.24	86.63 ± 0.61
C_{rounds}	18 ± 2	4 ± 1	28 ± 3	7 ± 2	54 ± 3	9 ± 2

Note: Data-FL represents the performance of CNN only, while XAI-FL represents the performance of CNN combined with TreeSHAP.

TABLE 5. The performance comparison between the data-based and XAI-based FL against attack #1 for Non-IID Scenario.

	$\alpha = 0.1$		$\alpha = 0.3$		$\alpha = 0.5$	
	Data-FL [24], [27], [48]	XAI-FL Proposed	Data-FL [24], [27], [48]	XAI-FL Proposed	Data-FL [24], [27], [48]	XAI-FL Proposed
ACC	97.94 ± 0.03	99.61 ± 0.08	88.60 ± 0.55	97.25 ± 0.09	77.64 ± 1.26	89.63 ± 0.16
PR	97.46 ± 0.14	99.50 ± 0.15	89.08 ± 1.91	97.19 ± 0.33	77.61 ± 2.51	89.13 ± 0.59
$F1$	97.96 ± 0.03	99.61 ± 0.08	88.52 ± 0.41	97.25 ± 0.08	77.65 ± 1.12	89.70 ± 0.12
DR	98.46 ± 0.15	99.72 ± 0.00	87.98 ± 1.12	97.30 ± 0.18	77.70 ± 1.83	90.28 ± 0.50
FA	2.57 ± 0.15	0.50 ± 0.15	10.79 ± 2.11	2.81 ± 0.35	22.42 ± 3.03	11.01 ± 0.73
HD	95.89 ± 0.06	99.22 ± 0.15	77.19 ± 1.10	94.49 ± 0.17	55.28 ± 2.53	79.27 ± 0.32
C_{rounds}	38 ± 1	8 ± 1	84 ± 1	38 ± 1	89 ± 2	41 ± 1

Note: Data-FL represents the performance of CNN only, while XAI-FL represents the performance of CNN combined with TreeSHAP.

TABLE 6. The performance comparison between the Data-based and XAI-based FL against *Attack* #2 for IID Scenario.

	$\alpha_t = [0, 0.5]$	
	Data-FL [24], [27], [48]	XAI-FL Proposed
ACC	96.54 ± 0.25	98.85 ± 0.18
PR	94.23 ± 0.35	98.39 ± 0.25
$F1$	96.63 ± 0.22	98.85 ± 0.20
DR	99.16 ± 0.10	99.33 ± 0.10
FA	6.07 ± 0.40	1.63 ± 0.15
HD	93.09 ± 0.30	97.70 ± 0.22
C_{rounds}	21 ± 4	3 ± 1

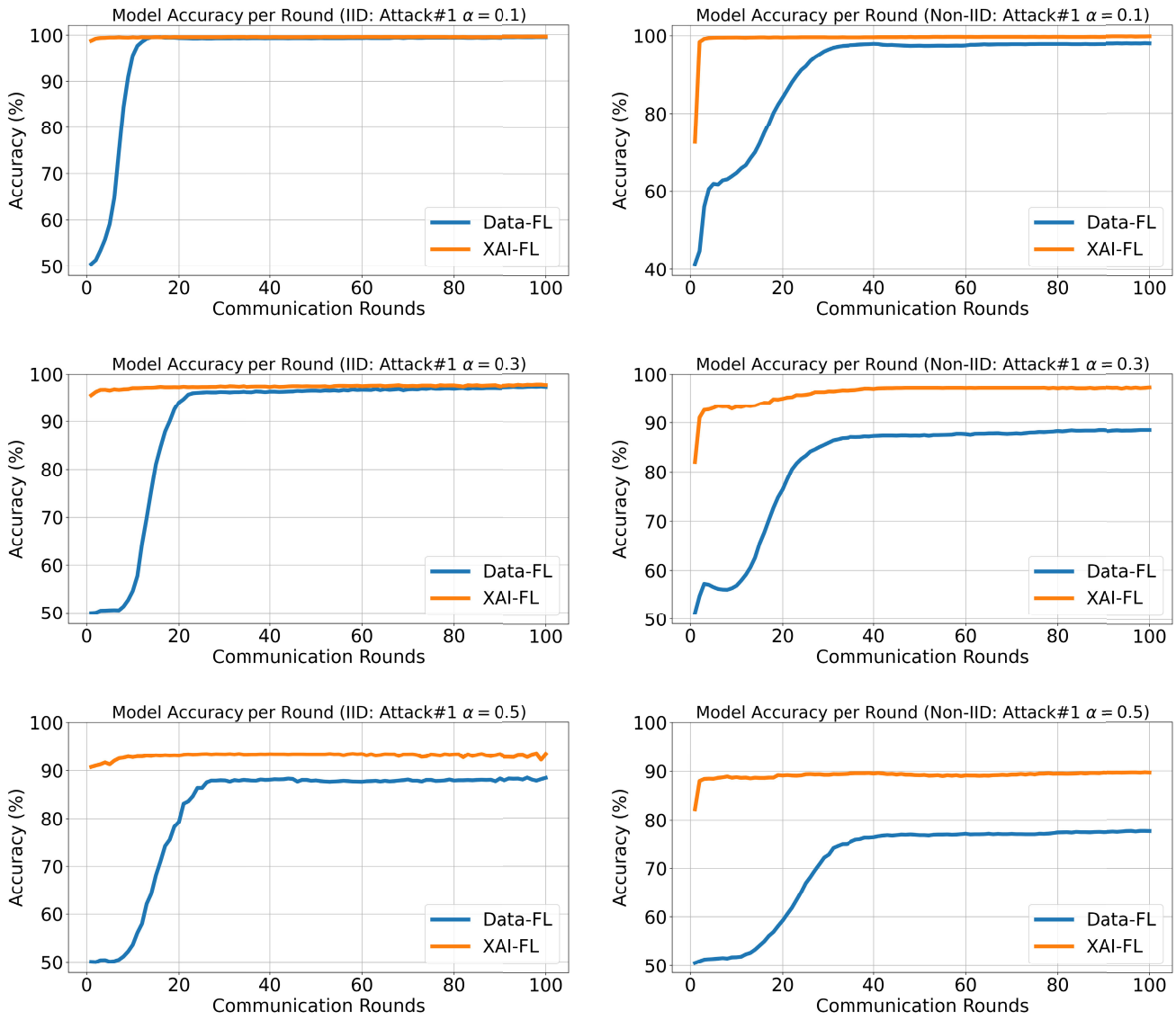
Note: Data-FL represents the performance of CNN only, while XAI-FL represents the performance of CNN combined with TreeSHAP.

(XAI-FL). FedAvg [48] is the original federated learning aggregation mechanism proposed by the Google team and is commonly applied in related works such as [24] and [27]. Each value in the description column gives the layer type, number of neurons, and activation function. Additionally, the Adam optimizer was employed to train our detectors with FL rounds = `100`, learning rate = `0.0001`, batch size = `50`, and loss function = `CategoricalCrossentropy`.

Table 3 presents the performance metrics based on the confusion matrix. In this matrix, True Positive (*TP*) refers to the number of malicious samples correctly classified, True Negative (*TN*) is the number of benign samples correctly classified, False Negative (*FN*) is the number of malicious samples incorrectly classified as benign, and False Positive (*FP*) is the number of benign samples incorrectly classified as malicious. These metrics are used to evaluate the effectiveness of our FL detectors, as commonly done in the literature. The metrics include Accuracy (*ACC*), Precision (*PR*), F1-score ($F1$), Detection Rate (*DR*), Precision (*PR*), and Highest Difference (*HD*). Additionally, Convergence Rounds (C_{rounds}) is used to evaluate the communication/computation overhead. Fewer number of convergence rounds indicates lower overhead, because fewer iterations are needed to achieve model convergence and compute the final model, thereby reducing the total time and resources required for training.

B. EXPERIMENTAL RESULTS

We conducted two experiments to compare the traditional data-based FL approach to our proposed XAI-based FL



(a) Attack #1 (IID Scenario).

(b) Attack #1 (Non-IID Scenario).

FIGURE 7. Accuracy over communication rounds in the data-based and XAI-based FL using ISET dataset.

approach. These comparisons were made in terms of ACC, PR, $F1$, DR, FA, HD, and C_{rounds} against various false data injection attacks outlined in Subsection IV-B, using both the ISET and PRECON datasets. Experiment #1 compares the performance using the ISET dataset, while experiment #2 compares the performance using the PRECON dataset. In both experiments, results were recorded over 10 runs, and variability is reported using standard deviation (stddev), expressed as $\pm stddev$.

1) EXPERIMENT #1

In this experiment, we trained two CNN-based FL detectors: one using raw power consumption readings (referred to as, data-FL) and the other using XAI explanations of power consumption readings (referred to as, XAI-FL). We evaluated the performance and overhead of the two approaches while analyzing their effectiveness against three different cyber-attacks (#1, #2, and #3). The detailed findings from this experiment are summarized as follows.

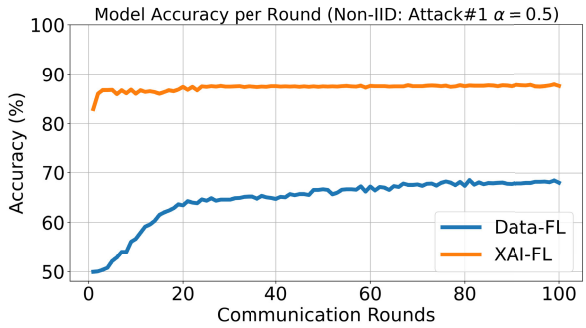
Table 4 presents the classification performance of data-FL and XAI-FL detectors against attack #1 using different values of α including $\{0.1, 0.3, 0.5\}$ for IID scenario. The results clearly demonstrate that the proposed XAI-FL approach improves classification performance metrics while reducing overhead by requiring fewer rounds to converge (C_{rounds}). Moreover, as α increases, indicating more stealthy attack, data-FL struggles to maintain performance, whereas XAI-FL performs significantly better. For example, under the most

TABLE 7. The performance comparison between the data-based and XAI-based FL against attack #2 for Non-IID Scenario.

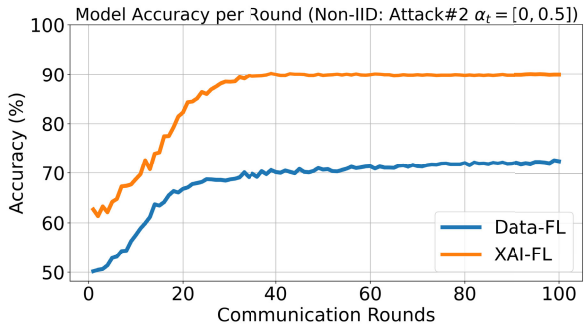
	$\alpha_t = [0, 0.5]$	
	Data-FL [24], [27], [48]	XAI-FL Proposed
ACC	90.79 ± 0.45	98.85 ± 0.20
PR	89.41 ± 0.85	98.12 ± 0.15
$F1$	90.94 ± 0.45	98.86 ± 0.20
DR	92.53 ± 1.35	99.61 ± 0.15
FA	10.96 ± 0.80	1.91 ± 0.25
HD	81.56 ± 0.60	97.70 ± 0.28
C_{rounds}	53 ± 2	18 ± 1

Note: Data-FL represents the performance of CNN only, while XAI-FL represents the performance of CNN combined with TreeSHAP.

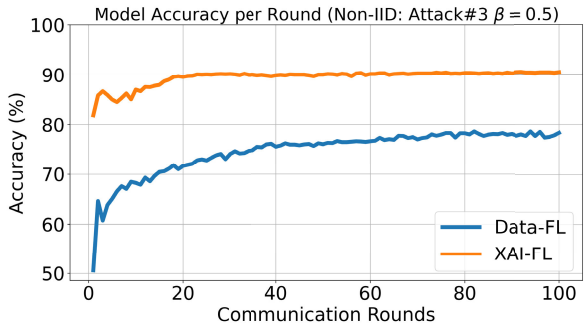
stealthy attack ($\alpha = 0.5$), the performance metrics (ACC, PR, $F1$, DR, FA, and HD) improve from (88.40, 86.61, 88.68, 90.84, 14.04, and 76.80) to (93.31, 93.91, 93.27, 92.64, 6.01, and 86.63), while the number of convergence rounds C_{rounds} decreases form 54 to 9. Table 5 presents results for the Non-IID scenario, showing similar trends. Here, the performance gap widens under the most stealthy attack ($\alpha = 0.5$). XAI-FL improves metrics like ACC and DR from (77.64 and 77.70) to (89.63 and 90.28) and reduces FA and C_{rounds} from (22.42 and 89) to (11.01 and 41), highlighting its efficiency. This improvement is due to XAI's ability to clearly distinguish between benign and malicious consumption patterns as shown in Figs. 4 and 5. This distinction streamlines the training process and enhances detection accuracy in XAI-FL compared to data-FL. In data-FL, training on raw data with complex, high-dimensional patterns requires more rounds to learn and produce a complex decision boundary to fit these data. In contrast, XAI simplifies the patterns, leading to a decision boundary that can be learned more effectively. Additionally, statistical significance testing using the t-test [49] was conducted to validate performance differences between data-FL and XAI-FL. Significant differences were observed across all metrics. For example, under the most stealthy attack ($\alpha = 0.5$) in the IID scenario, the p-values of $ACC = 0.0041$, $PR = 0.0001$, $F1 = 0.0095$, $DR = 0.0155$, $FA = 0.0000$, and $HD = 0.0041$. Moreover, in terms of ablation comparison, it is worth noting that Data-FL represents CNN-only model, while XAI-FL represents CNN combined with TreeSHAP. Therefore, the performance difference between XAI-FL and Data-FL reflects the contribution of TreeSHAP—for example, for Attack #1 when $\alpha = 0.5$, the contribution of XAI is an improvement of approximately 4.91 ($93.31 \pm 0.30 - 88.40 \pm 1.12$) for IID and 11.99 ($89.63 \pm 0.16 - 77.64 \pm 1.26$) for Non-IID in terms of ACC, and a reduction of approximately 45 ($54 \pm 3 - 9 \pm 2$) for IID and 48 ($41 \pm 1 - 89 \pm 2$) for Non-IID in terms of C_{rounds}, as shown in Tables 4 and 5. This pattern also applies to other attacks (#2 and #3).



(a) *Attack* #1.



(b) *Attack* #2.



(c) *Attack* #3.

FIGURE 8. Accuracy over communication rounds in the Data-based and XAI-based FL using PRECON dataset.

Tables 6 and 7 present the classification performance of Data-FL and XAI-FL detectors in case of *Attack* #2 for IID and Non-IID scenarios, respectively. The results clearly show that the proposed XAI-FL approach enhances the classification performance metrics (ACC, PR, $F1$, DR, FA, and HD) in both IID and Non-IID scenarios, and reduces C_{rounds} from 21 to 3 in IID and from 53 to 18 in Non-IID.

Table 8 presents the classification performance of data-FL and XAI-FL detectors in case of attack #3 using different values of α for IID scenario. The results clearly demonstrate that the proposed XAI-FL approach improves classification

TABLE 8. The performance comparison between the data-based and XAI-based FL against attack #3 for IID Scenario.

	$\beta = 0.1$		$\beta = 0.3$		$\beta = 0.5$	
	Data-FL [24], [27], [48]	XAI-FL Proposed	Data-FL [24], [27], [48]	XAI-FL Proposed	Data-FL [24], [27], [48]	XAI-FL Proposed
ACC	99.61 ± 0.15	99.78 ± 0.10	95.73 ± 0.45	97.42 ± 0.13	85.70 ± 0.90	93.03 ± 0.45
PR	99.66 ± 0.12	99.66 ± 0.05	96.25 ± 0.40	96.58 ± 0.10	86.21 ± 0.95	91.77 ± 0.20
$F1$	99.61 ± 0.14	99.78 ± 0.10	95.71 ± 0.50	97.44 ± 0.11	85.60 ± 1.10	93.14 ± 0.30
DR	99.55 ± 0.20	99.89 ± 0.05	95.17 ± 0.55	98.31 ± 0.20	85.00 ± 1.20	94.55 ± 0.17
FA	0.34 ± 0.08	0.34 ± 0.03	3.71 ± 0.35	3.48 ± 0.30	13.60 ± 1.50	8.48 ± 0.14
HD	99.21 ± 0.25	99.55 ± 0.15	91.46 ± 0.80	94.83 ± 0.14	71.40 ± 2.50	86.07 ± 0.34
C_{rounds}	21 ± 3	6 ± 1	64 ± 4	14 ± 2	97 ± 1	37 ± 2

Note: Data-FL represents the performance of CNN only, while XAI-FL represents the performance of CNN combined with TreeSHAP.

TABLE 9. The performance comparison between the data-based and XAI-based FL against attack #3 for Non-IID Scenario.

	$\beta = 0.1$		$\beta = 0.3$		$\beta = 0.5$	
	Data-FL [24], [27], [48]	XAI-FL Proposed	Data-FL [24], [27], [48]	XAI-FL Proposed	Data-FL [24], [27], [48]	XAI-FL Proposed
ACC	98.65 ± 0.25	99.78 ± 0.10	90.98 ± 1.25	97.64 ± 0.30	81.99 ± 1.50	92.33 ± 0.41
PR	97.53 ± 0.30	99.72 ± 0.15	87.62 ± 1.40	96.85 ± 0.31	78.75 ± 1.60	91.02 ± 0.50
$F1$	98.67 ± 0.22	99.78 ± 0.05	91.37 ± 0.15	97.66 ± 0.14	82.96 ± 1.55	92.45 ± 0.45
DR	99.83 ± 0.10	99.83 ± 0.10	95.45 ± 0.80	98.48 ± 0.26	87.64 ± 1.20	93.93 ± 0.20
FA	2.53 ± 0.45	0.28 ± 0.07	13.48 ± 2.60	3.20 ± 0.25	23.65 ± 2.50	9.27 ± 0.62
HD	97.30 ± 0.35	98.55 ± 0.28	81.97 ± 2.20	95.28 ± 0.55	63.99 ± 3.10	84.66 ± 1.01
C_{rounds}	64 ± 4	11 ± 2	86 ± 3	23 ± 1	98 ± 2	34 ± 2

Note: Data-FL represents the performance of CNN only, while XAI-FL represents the performance of CNN combined with TreeSHAP.

TABLE 10. The performance comparison between the data-based and XAI-based FL using PRECON dataset.

	Attack #1		*Attack #2*		*Attack #3*	
	Data-FL [24], [27], [48]	XAI-FL Proposed	Data-FL [24], [27], [48]	XAI-FL Proposed	Data-FL [24], [27], [48]	XAI-FL Proposed
ACC	68.04 ± 2.20	87.42 ± 1.40	72.41 ± 1.80	89.81 ± 0.45	78.33 ± 0.35	90.40 ± 0.03
PR	66.53 ± 2.50	89.15 ± 1.20	70.96 ± 2.00	90.25 ± 0.40	78.80 ± 0.40	90.37 ± 0.04
$F1$	69.45 ± 2.10	87.14 ± 1.50	73.32 ± 1.70	89.75 ± 0.50	78.15 ± 0.45	90.41 ± 0.03
DR	72.64 ± 1.80	85.22 ± 1.01	75.85 ± 1.90	89.26 ± 0.55	77.50 ± 0.65	90.45 ± 0.05
FA	36.55 ± 3.50	10.38 ± 1.70	31.04 ± 2.50	9.64 ± 1.20	20.84 ± 1.50	9.64 ± 0.25
HD	36.09 ± 3.70	74.84 ± 1.81	44.81 ± 3.00	79.62 ± 1.80	56.66 ± 1.95	80.81 ± 0.55
C_{rounds}	84 ± 3	29 ± 2	88 ± 2	38 ± 2	92 ± 2	25 ± 2

Note: Data-FL represents the performance of CNN only, while XAI-FL represents the performance of CNN combined with TreeSHAP.

performance metrics while reducing overhead by requiring fewer rounds to converge (C_{rounds}). Moreover, as α increases, indicating more stealthy attack, data-FL struggles to maintain performance, whereas XAI-FL performs significantly better. For example, under the most stealthy attack ($\alpha = 0.5$), the performance metrics (*ACC*, *PR*, $F1$, *DR*, *FA*, and *HD*) improve from (85.70, 86.21, 85.60, 85.00, 13.60, and 71.40) to (93.03, 91.77, 93.14, 94.55, 8.48, and 86.07), while the number of convergence rounds decreases form 97 to 37. Table 9 presents the results of the Non-IID scenario, showing similar trends. Here, the performance gap widens under the most stealthy attack ($\alpha = 0.5$). XAI-FL improves metrics like *ACC* and *DR* from (81.99 and 87.64) to (92.33 and 93.93) and reduces *FA* and C_{rounds} from (23.65 and 98) to (9.27 and 34), highlighting its efficiency. When comparing attacks #1 and #2 to attack #3, it is evident that attack #3 is more challenging to detect due to its intermittent behavior, which makes malicious data resemble benign data. Consequently, data-FL takes

more rounds to converge, while XAI-FL converges more efficiently in fewer number of rounds. This is because XAI can effectively distinguish between benign and malicious consumption patterns, making the training process smoother and enhancing detection.

Fig. 7 provides a visual representation of accuracy over communication rounds for both IID and Non-IID scenarios. It can be seen that the proposed XAI-FL approach outperforms data-FL in attack #1, requiring fewer number of communication rounds to converge. This finding is also valid for attacks #2 and #3, but due to space limitations, only the results of attack #1 are included.

2) EXPERIMENT #2

In this experiment, we trained two CNN-based FL detectors (data-FL and XAI-FL) using the PRECON dataset to assess their applicability and effectiveness across different data sources. We evaluated the performance and overhead of the detectors while examining their effectiveness against three distinct cyber-attacks (#1, #2, and #3) at high values of α, α_t, and β to emulate stealthy attacks. The detailed findings from this experiment are summarized in Table 10 and Fig. 8, highlighting the key results as follows.

Table 10 presents the classification performance of data-FL and XAI-FL detectors in case of attacks #1, #2, and #3. The results demonstrate that the proposed XAI-FL outperforms data-FL in terms of ACC, PR, $F1$, DR, FA, HD, and C_{rounds}. For example, the proposed XAI-FL approach improves metrics like ACC and DR from (68.04 and 72.64) to (87.42 and 85.22) in attack #1, from (72.41 and 75.85) to (89.81 and 89.26) in attack #2, and from (78.33 and 77.50) to (90.40 and 90.45) in attack #3. Additionally, it reduces FA and C_{rounds} from (36.55 and 84) to (10.38 and 29) in attack #1, from (31.04 and 88) to (9.64 and 38) in attack #2, and from (20.84 and 92) to (9.64 and 25) in attack #3. These results in addition to the results of experiment #1 highlight the proposed XAI-FL's ability to perform effectively across different datasets with improved performance and reduced overhead, thanks to its XAI capabilities. Additionally, Fig. 8 provides a visual representation of accuracy over communication rounds. It can be seen that the proposed XAI-FL approach outperforms data-FL in attacks #1, #2, and #3, requiring fewer number of communication rounds to converge.

VI. CONCLUSION AND FUTURE WORKS

This paper addresses two key challenges in the FL training process: lower performance compared to CT methods and higher overhead compared to existing data-based FL approaches. To explore these issues, we used the detection of false data in SGs as a case study. To tackle these challenges, we proposed an XAI-based FL approach. Unlike traditional data-based FL methods, our approach trains on the explanations of electricity consumption data rather than raw data. This idea not only enhances detection accuracy but also significantly reduces overhead by minimizing the number of FL rounds required for convergence. As a result, the proposed method offers a more efficient and effective solution for decentralized learning. Experimental evaluations on two real-world datasets, ISET and PRECON, under various cyber-attack scenarios, demonstrated the superiority of the XAI-based FL approach. It consistently outperformed traditional data-based FL methods in both accuracy and computational overhead, especially in detecting stealthy false data injection attacks. For instance, in some cases, our approach was able to reduce the number of FL rounds by more than 80%, reduce the false alarm by more than 82%, and increase the detection accuracy by more than 14%. These findings underscore the potential of XAI-driven strategies to improve the efficiency, scalability, and performance of FL training processes.

This paper primarily examined the performance benefits of the XAI-FL approach compared to data-FL in detecting false data. As part of future research, we aim to explore how both approaches respond to adversarial threats, particularly evasion, model inversion, and backdoor attacks. These attacks represent significant challenges in machine learning security. Evasion attacks occur during model inference, where an attacker introduces small, often imperceptible, changes to input data to mislead the model into making incorrect decisions. Since XAI-FL leverages explanation-based features, which exhibit greater class separation than raw data, we anticipate that evasion attacks may require larger perturbations to succeed—potentially making XAI-FL more resilient than data-FL. Model inversion attacks attempt to reconstruct sensitive training data. On the other hand, backdoor attacks target the training phase by embedding hidden triggers into the data, causing the model to output a predefined label whenever the trigger is present. Both attack types can severely compromise model integrity and pose significant risks, especially in critical systems. It would also be valuable to explore how different explanation computation methods affect model accuracy, computational efficiency, and resilience to adversarial attacks.

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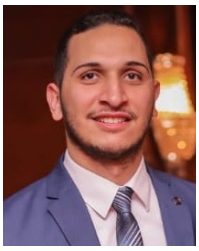

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